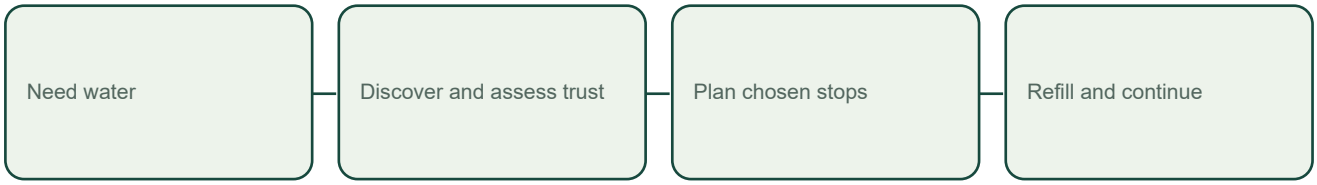
AlunUÉ
Water along the way

Exploratory research and requirements for refill discovery and student walking journeys in Évora.

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English review copy. The Portuguese PDF is the intended coursework version. This copy explains the same evidence, design proposals and limitations for the author to review.

Open the app · GitHub Pages | Repository and checks



Conceptual task diagram. Not an observed journey, street map or verified water directory.

Scope and course fit

The proposal is an essential-services and mobility module within AlunUÉ: find trustworthy information about a place to refill a bottle and continue the student day. Supermarkets, cultural stops and willing shops are optional extensions. Phase 1 focuses on research and requirements; a working prototype supports exploration and does not establish that the concept has been validated.

State of the work

This is individual work. The author says the lecturer approved individual work; documentary approval is not presented. Available research consists of recalled informal conversations. The three personas and maps are provisional and require validation. No recoverable structured survey or usability study has been completed. Evidence, interpretation and proposed design are kept distinct; missing data is not replaced by fabricated findings.

Method and exploratory signals

Source and conditions of collection

The author recalls discussing an app or website idea with acquaintances, including Erasmus students and people in his faculty. He identifies them as university students; each person's UÉ enrolment was not independently checked. About 30 people is an approximate memory, not an auditable participant count. The original notes are unavailable; the author reports their loss. Dates, profile distribution and complete individual responses cannot be reconstructed.

The remembered question was approximately whether a service like this would be useful. Conversations were casual, and the idea was presented before the reaction. This explores stated acceptance but introduces convenience, confirmation and social-desirability biases. It does not establish frequent need, adoption or repeated use.

ID / category	Content and evidential limit
E1 · author recollection	People reacted positively to the idea; some mentioned buying water repeatedly. No exact counts, transcripts or complete individually attributed responses were recovered.
E2 · author recollection	The author recalls that people did not know refill points existed or where to find them. The prevalence of this lack of awareness was not measured.
E3 · author preference	Trust, water discovery, walking routes, groceries along the way, views and participating shops are design priorities. They are not survey rankings.
E4 · author confirmation	No participating shop has been found yet. A shop network is a future proposal, with no agreement or permission to enter established.

Interpretation and participant protection

E1 and E2 suggest investigating discovery and trust. They do not support percentages, segment comparisons or causal conclusions. No individual objections were reconstructed; missing objections do not prove that none existed. Interlocutors are not named. Consent for academic use was not recovered, and private messages are not published. Future interviews should obtain permission for an anonymous summary, collect only necessary information and allow withdrawal.

Three working personas

These are proposed proto-personas, with no invented interviewee names or biographies. Only the Erasmus context has an explicit connection to the author's reported conversations. The other profiles are recruitment hypotheses suggested by the brief, not segments already studied.

Profile	Goal, context and required validation
P1 · Newly arrived international student	Goal: find water without knowing the city or speaking fluent Portuguese. Possible barriers: vocabulary, trust and discomfort entering a shop. The Erasmus context is recalled; the specific barriers remain hypotheses. Validate through the person's last real experience of looking for water.
P2 · First-year student	Goal: combine a refill with travel between classes. Possible barriers: unfamiliar buildings and unclear opening hours. No specific evidence for this segment is available. Validate knowledge of spaces, access and current alternatives.
P3 · Working student	Goal: meet the need with little detour and possibly combine a purchase. Possible constraints: limited time and low tolerance for stale information. No specific evidence for this segment is available. Validate time constraints, decisions and problem frequency.

Turning hypotheses into research-grounded personas

Recruit real people across these contexts, confirm their UÉ connection, ask about concrete experiences before showing the idea, record both agreements and disagreements, and remove unsupported attributes. Do not assume age, income, nationality or digital skills explain behaviour.

Proposed follow-up interview guide

1. Tell me about the last time you needed water while out in Évora. 2. What did you do and why? 3. How did you know you could use the place? 4. Was there a cost or detour? 5. Have you asked a shop for water? 6. What information was missing? 7. What would make you avoid this solution? 8. May an anonymous summary be used for coursework?

Record an anonymous code, actual date, context, declared academic connection, question, paraphrase and confidence. A new interview keeps its real date and cannot retroactively replace lost notes.

Empathy maps

These maps express design hypotheses, not observed mental states. “Says / goal” is a project formulation, without quotation marks or attribution to an interviewee.

P1 · International student

Dimension	Hypothesis to validate
Says / goal	Find somewhere to fill a bottle.
Thinks	Is the water safe and free? Can I ask for help?
Does	Search for information and compare alternatives.
Feels	Possible uncertainty in an unfamiliar place.

P2 · First-year student

Dimension	Hypothesis to validate
Says / goal	Understand where water is available between classes.
Thinks	Will the place be accessible before the next class?
Does	Plan stops near the usual route.
Feels	Possible frustration if a point is closed.

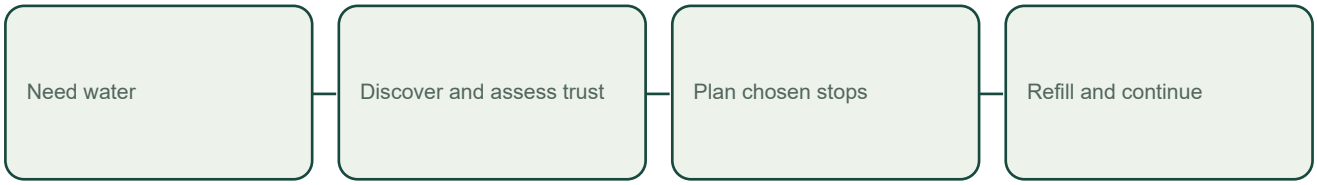
P3 · Working student

Dimension	Hypothesis to validate
Says / goal	Solve the need without a large detour.
Thinks	Can water and a purchase be combined?
Does	Compare alternatives and adjust stop order.
Feels	Possible concern about delays and stale information.

Questions still to investigate: what increases trust? Is entering a shop uncomfortable? What mobility, language, opening-hour or connectivity barriers exist? No recorded answers to these questions are claimed.

Proposed user journey

Design scenario: a student needs to refill a bottle while travelling through the city. This sequence has not been observed in a field study or usability test.



Conceptual task diagram. Not an observed journey, street map or verified water directory.

Stage	Action and possible difficulty	Design response
1 · Need	Look for water without knowing where to start.	Searchable list, optional location and manual start.
2 · Assess	Distinguish a useful point from uncertain map data.	Source, date, access, cost and verification status.
3 · Plan	Choose stops without an unnecessary detour.	Adjustable order and optional grocery/cultural stops.
4 · Travel	Move the plan to external navigation.	Walking directions export and a text alternative.
5 · Arrive	Encounter a closure or incorrect information.	Show limits, seek an alternative and prepare a correction.
6 · Continue	Continue without pressure to spend or interact.	No purchase requirement; shops only with explicit opt-in.

Profile variations

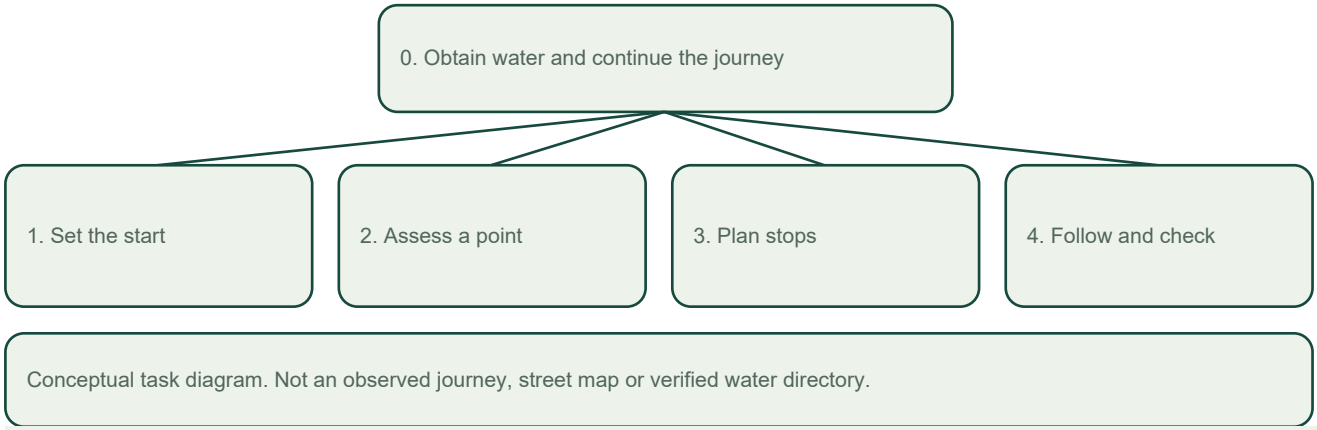
P1: clear language and guidance for approaching a shop; Portuguese use should be tested with international students. P2: building access and hours matter. P3: control over stops and detours matters. These differences are hypotheses for future interviews and tests.

Physical mapping and geographic limits

The prototype can open an Évora street map and request OpenStreetMap candidates. Synthetic coordinates do not identify real water services. No verified fountain map is included because a field inventory has not been completed. A mapped point does not establish potability, free access or operational status.

Hierarchical task analysis

Goal 0: obtain water and continue the journey. This HTA describes intended tasks and recovery paths, rather than behaviour already observed.



Task	Subtasks and completion conditions
1 · Set a start	1.1 Choose centre/campus or coordinates. 1.2 Request location only if wanted. 1.3 Keep a manual option when permission is refused.
2 · Assess a point	2.1 Inspect the list. 2.2 Read source and date. 2.3 Check water suitability, cost, access and operation. 2.4 Reject unconfirmed candidates from a real water journey.
3 · Plan stops	3.1 Add an eligible point. 3.2 Optionally add a supermarket or cultural stop. 3.3 Reorder. 3.4 Review. 3.5 Export in parts or save text.
4 · Follow and check	4.1 Consult external navigation. 4.2 Confirm conditions at the location. 4.3 Refill if allowed. 4.4 Choose an alternative and record a suggestion if information fails.

Plans and exceptions

Main plan: 1 → 2 → 3 → 4. Repeat task 2 if no eligible point is available. Without internet, show a dated stored snapshot and saved itinerary; do not promise offline street maps. Without location, use task 1.1. If external services fail, preserve the list and offer text. A shop without agreement must not be shown as a participating partner.

Operations explored in the prototype

Type filters, search, an identified demonstration mode, selected and ordered stops, walking export, share/copy with text-file fallback, device-only suggestion drafts and optional service requests. There is no verified operational directory, moderation service or real partner registration.

Requirements and evaluation criteria

Project priorities: P0 essential; P1 extension. Targets below are proposed criteria for future testing, not measured results.

ID / priority	Requirement and traceability	Proposed criterion
R1 · P0	Distinguish candidates, demonstrations and confirmed points. E2 plus misinformation risk.	Never describe an unverified candidate as guaranteed drinking water.
R2 · P0	Find water without an account or mandatory location. E1/E2 and accessibility hypothesis.	At least 4 of 5 participants complete a manual-start task.
R3 · P0	Show source, date, cost and access. E2 and trust decision.	At least 4 of 5 correctly identify uncertainty before choosing.
R4 · P0	Preserve all stops on export. E3.	No chosen stop is lost; a readable text alternative exists.
R5 · P0	Recover from network/API failure. Robustness decision.	The plan is preserved and the failure explained.
R6 · P1	Combine optional grocery/cultural stops. E3.	Each extension can be excluded while keeping water as the focus.
R7 · P1	Voluntary shops without mandatory purchase. E3/E4.	No shop is listed as a partner without explicit agreement.

Usability and accessibility

Design goals: European Portuguese labels, visible focus, keyboard access, readable contrast, zoom without content loss, suitable touch controls and recovery messages. Evaluate status recognition, completion, errors, task time and satisfaction. No SUS result was collected in this phase.

Information architecture

Discover: search, filters and sources. My journey: start, stops and export. Community: opt-in proposal and local drafts. The project: evidence, limits and sources. Planning a journey does not require joining the shop network.

Limits, next steps and sources

Limits affecting this phase

The research is exploratory and retrospective. Complete individual records, a structured questionnaire, a verified participant population and validation of the three personas are missing. A positive reaction to a presented idea does not measure adoption. The design artefacts must therefore not be treated as the results of a completed user study. Losing the notes explains the limitation but does not remove it.

Step	Outcome to obtain
1 · Further research	Neutral interviews with real students; establish needs before showing the concept and preserve authorised anonymous summaries.
2 · Revise Phase 1	Update personas, maps and requirements from collected evidence; keep a change record.
3 · Evaluate prototypes	Paper and medium-fidelity tasks, followed by heuristic evaluation required by the course.
4 · Final phase	At least five real UÉ users, task metrics and SUS; report only actual observations.
5 · Operational service	Verify water points in the field, assign maintenance and obtain shop agreement. Test the Android PWA before deciding on APK distribution.

Use of generative AI

Generative AI assisted report structure and Portuguese drafting, design artefacts, diagrams, prototype code and translation. The author supplied recollections and scope decisions. AI did not conduct interviews or verify places. The author should review and understand the content and remains responsible for the authenticity of the submission.

Sources and supporting materials

1. INF13186L, Introdução à UC, pp. 5-6: AlunUÉ scope, ten-page report limit and AI disclosure.
2. INF13186L, gemini_fase1, p. 1: research, personas, empathy/journey maps and HTA. Later example pages were not used as interview evidence.
3. Author statements and recollections, 30 October 2026.
4. OpenStreetMap attribution; official Google Maps URLs documentation.
5. Public app and dedicated repository. The PDF stands on its own; links do not replace the required design artefacts.